

Stylized Facts about AI & Human Capabilities

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We discuss stylized facts about the capabilities of recent AI models and how they compare to human capabilities. We particularly draw attention to a common latent factor of capabilities among AI models, and the fact that this latent factor is different from the distribution of human capabilities. We discuss various candidates for the “missing capability” which would unlock a greater economic impact.

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This document attempts to summarize facts about AI and human capabilities relevant for understanding the economic impacts of AI.

This document does not contain non-public information about the capabilities of OpenAI’s models or our research directions.

None of the statements reflect the views of OpenAI.

1 Overview

1.1 We want to characterize AI capabilities

The goal of this note is to give a crisp characterization of AI capabilities: (1) what tasks can current AI models do?; (2) how do different AI models differ?; (3) how do AI abilities differ from human abilities?; (4) what are reasonable expectations about future AI abilities?

A running theme will be that the characterization of AI capabilities is *difficult*: there is a long history of struggle in making accurate generalizations about the capabilities of artificial intelligence. There is little consensus on how to describe the abilities of contemporary models.

In this note we list a series of well-documented facts about AI capabilities, and make some more opinionated generalizations. The target audience is economists who are interested in modeling the effects of AI on various situations.

1.2 We wish to identify latent factors of ability

It is useful to think of a matrix of agents and tasks.

Suppose we can observe whether an agent can perform a task, for a long list of agents and tasks. The agent could be a human, a computer, or a hybrid. We can represent each observation in a matrix with agents and columns and tasks as rows:

	Alice	Bob	Charlie	AI in 2020	AI in 2024	AI in 2028
What's 2+2?						?
How many moons on Neptune?						?
Pass LSAT						?
Complete Super Mario Bros						?
Prove Reimann hypothesis						?

In practice the matrix is enormous. We can think of the capability problem as finding a lower dimensional representation of this matrix, i.e. some small set of general-purpose capabilities that are highly predictive of whether an agent can do a task.¹

The simplest factorization is to assign each agent a single score representing their ability, and to assign each task a score representing its difficulty. As we will see below a single-factor representation of ability is often a poor fit, and it would be practically very useful to have 2 or 3 factors to help understand the nature of capabilities and tasks.

1.3 There is no standard measure of machine intelligence

The problem of defining machine intelligence is notoriously difficult. Many tests or tasks have been proposed as benchmarks of intelligence but they have been consistently disappointing: when computers pass the test we find they still don't exhibit general-purpose intelligence.

The table in the margin shows a sparse history of when computers could achieve various .

Scott Alexander [says](#) “The history of AI is people saying ‘We’ll believe AI is Actually Intelligent when it does X!’ – and then, after AI does X, not believing it’s Actually Intelligent.”

1.4 There is no standard taxonomy of AI capabilities.

There are hundreds of different AI benchmarks of capability. People often group those benchmarks informally into groups, e.g. “agentic”, “reasoning”, “general knowledge”, however there is no generally-accepted taxonomy of abilities.

Some papers have proposed canonical taxonomies but none have yet been widely adopted:

¹We can think of the reduction either as Principal Components Analysis (we take weighted averages of all rows), or Principal Feature Analysis (we take a subset of rows that are highly predictive).

Benchmark
Arithmetic
Chess
Jeopardy
Go
¹ DoTA
Winograd Schemas - “when people pass the test, we were thinking.”
MMMU – “Benchmark for Expert AGI.”
ARC-AGI – “the only formal benchmark of AGI progress.”
AGIEval - “general abilities ... in tasks pertinent to human intelligence and problem-solving.”

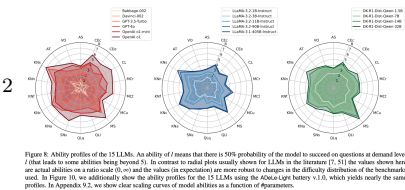
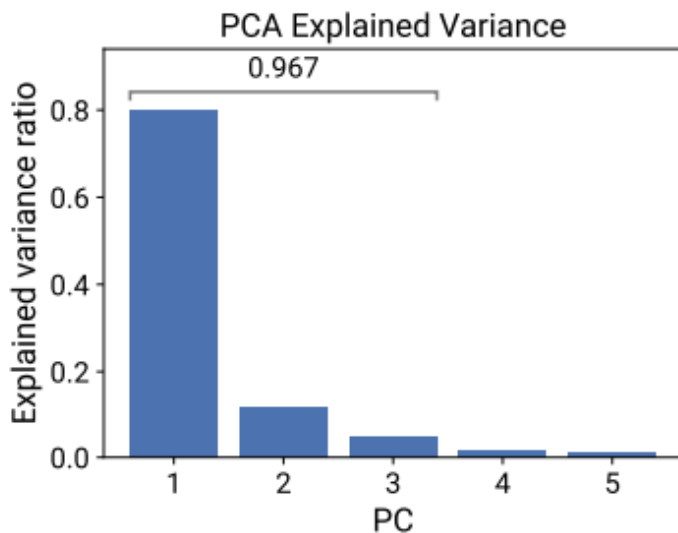


Figure 8: Ability profiles of the 15 LLMs. An ability of 1 means that there is 50% probability of the model to succeed on questions at demand level 1 (that leads to some abilities being harder). In contrast to valid plots usually shown for LLMs in the literature [7, 51] the values shown here are actual abilities on a ratio scale (0, ∞) and the values (on expectation) are more robust to changes in the difficulty distribution of the benchmarks used. In Figure 10, we additionally show the ability profiles for the 15 LLMs using the official open history 1.0, which yields nearly the same profiles. In Appendix 9.2, we show clear scaling curves of model abilities as a function of parameters.

- Lexin Zhou et al. (2025) propose an index of 18 different human-interpretable capabilities. They use an LLM to label hundreds of tasks from benchmarks according to the demand for each capability thereby allowing for the prediction of LLM capabilities on new benchmarks or tasks.
- Huber and Niklaus (2025) organize benchmark tasks according to “Bloom’s Taxonomy”, a heirarchical framework designed to classify educational learning objectives for humans, with 6 levels: Remember, Understand, Apply, Analyze, Evaluate, Create. They say there are a lack of benchmarks to measure the two highest levels (“evaluate” and “create”).

1.5 There does appears to be a single latent component of intelligence across LLMs

We discuss these facts in more detail below.



(a) PCA explained variance

Figure 1: From Ruan, Maddison, and Hashimoto (2024)

Historically, the AI capabilities distribution was very varied. For most of the history of AI specialized systems were built for different tasks, thus DeepBlue was good at Chess and Watson was good at Jeopardy, but each of them was bad at the other task.

LLM abilities are highly correlated across models. LLM performance is very highly correlated across benchmarks: if model A beats model B on benchmark X, then it very likely also beats model B on benchmark Y. We discuss some literature below which tries to quantify the strength of this correlation.

AI researchers treat intelligence as a scalar. A common sentiment among AI researchers is that it is best to spend time on making a model generally better, rather than fine-tuning for a specific case. This sentiment famously named the “bitter lesson” by Sutton (2019):

“general methods that leverage computation are ultimately the most effective, and by a large margin.”

Work on “scaling laws” treats model intelligence as a scalar. A large part of academic and industry work on AI is spent estimating “scaling laws”, i.e. predicting model performance based on various inputs (e.g. training compute, training data size, model scale, and test-time compute). Most work on scaling laws uses a scalar outcome to represent model intelligence, e.g. prediction error on a corpus, or performance on a difficult benchmark. In principle scaling laws could estimate a *vector* of intelligence outputs however this is relatively rare, researchers tend to treat a single scalar as a sufficient proxy for model intelligence.²

1.6 Latent LLM intelligence appears to be different from latent human intelligence

Humans have a fairly positive correlation among abilities.

- There is a reasonable correlation among different measures of general ability (IQ, g-factor).
- There is a reasonable correlation between years of education, IQ, standardized tests score, and wage.

Ideally we would compare human and LLM performance on the same tasks.

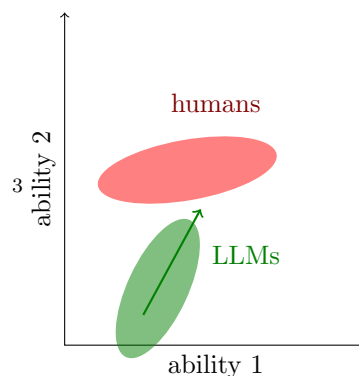
- If we had a full matrix of human and LLM performance we would be able to compare latent factors of each. Unfortunately there are few datasets which collect human performance across a range of distinct benchmarks.

We can compare the average human with the average LLM.

- Many benchmarks do report performance for the average or expert human, so we can make those comparisons. Additionally we can run LLMs on many human standardized tests and exams.

Latent factors for human and LLM ability appear to be significantly different.

- Aptitude tests that are predictive for humans are not predictive for LLMs. GPT-4 was able to get human-level scores at most aptitude tests: SAT, LSAT, civil service. But GPT-4 did not have the aptitudes passing such tests implies for humans.
- LLMs are able to do tasks no human can: LLMs can translate between pairs of languages which no human knows (Aymara to Kazakh), can combine knowledge from two very different domains.



²Polo et al. (2024) propose a decomposition with three latent factors: reasoning, knowledge, and instruction-following.

1.7 There is *some* bottleneck on LLMs impact, but disagreement about where

In 2023 many people thought GPT-4 could do a significant fraction of work:

- Eloundou et al. (2023): estimated that GPT-4 would reduce the time required by at least 50% for between 14% and 56% of all tasks in the economy.
- Chui et al. (2023) (McKinsey): *“the total percentage of hours that could theoretically be automated by integrating technologies that exist today ... [is] 60–70 percent.”*

In retrospect this seems to have been overly optimistic: GPT-4 performs very well on tests but there’s something about real-world tasks that trip it up.

[Andrej Karpathy](#), Nov 2024:

“Even though by many accounts (evals), LLMs are inching well into top expert territory (e.g. in math and coding etc.), you wouldn’t hire them over a person for the most menial jobs. It’s an interesting challenge how we can create evals for all the “easy” stuff that is secretly hard. Very long context windows, coherence, autonomy, common sense, multi-modal I/O that works, ... How do we build good “menial job” evals?”*

[Shunyu Yao](#), April 2025:

“AI has beat world champions at chess and Go, surpassed most humans on SAT and bar exams, and reached gold medal level on IOI and IMO. But the world hasn’t changed much, at least judged by economics and GDP. I call this the utility problem, and deem it the most important problem for AI.”*

1.8 There are many candidates for the bottleneck

We can ask why LLMs score well on aptitude tests but do not have the aptitude for real-world tasks:

Bottleneck	Description
Novel reasoning	Questions on aptitude tests are usually drawn from a standard pool, and don't involve truly novel chains of deduction.
Long context	Questions on aptitude tests have short prompts, real-world problems involve rich context.
Self-awareness	Humans have tacit knowledge of what is appropriate in a situation, not included on aptitude tests.
Accuracy and calibration	Aptitude tests reward guessing.
Qualitative outputs	Questions on aptitude tests have right or wrong answers, real-world work requires creative constraint satisfaction.
Interactive exploration	Questions on aptitude tests do not require a series of decisions, navigating an uncertain situation.
Long horizon tasks	Aptitude tests answers are immediately verifiable, many real-world decisions have long-delayed consequences.

2 LLM Performance on Tasks

In this section we discuss performance on tasks that have a measurable output.

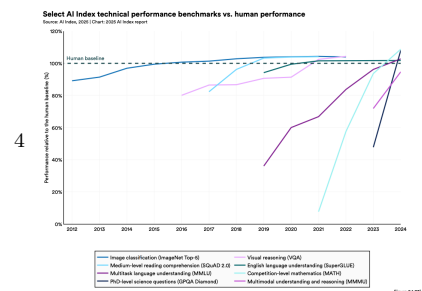
2.1 The time between benchmark introduction and saturation has been decreasing

If we had a cardinal scale of computer intelligence we could talk about the rate of improvement in a precise way.

The best proxy for velocity is the history of AI benchmarks: new benchmarks are introduced when computers achieve human-level performance on the old benchmarks (“saturation”).

Since 2012 the rate of benchmark saturation seems to be increasing:

Note that the inputs are growing very rapidly: “effective scale” has been growing around 15X/year.



2.2 LLM performance is highly correlated across benchmarks

Frontier models that surpass previous models on one benchmark usually outperform them on most other benchmarks.

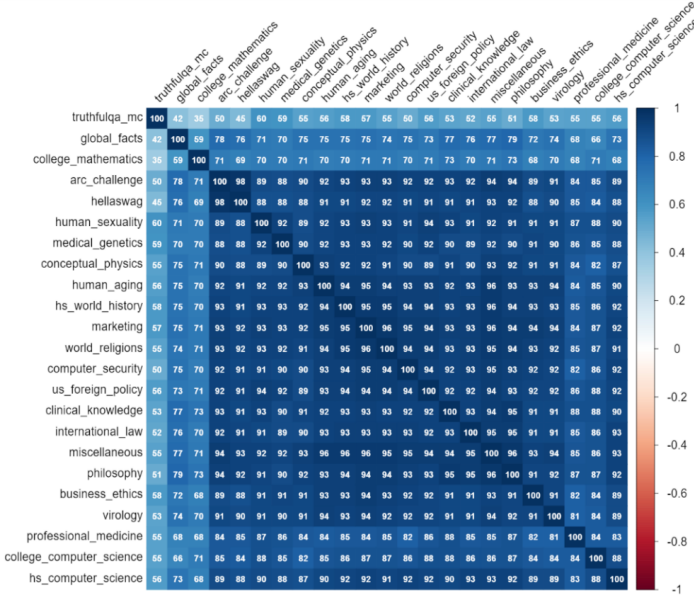


Fig. 3. Open LLM Leaderboard correlation matrix

Figure 6: Pairwise Correlations between benchmarks listed in LLMs.

Figure 2: From Spangher et al. (2024)

- Burnell et al. (2023) look at variation across 29 LLMs and 27 tasks, find 3 factors representing “reasoning, comprehension and core language modeling.” The models are mostly from 2022, i.e. prior to GPT-4.
- Ilić and Gignac (2024) tested over 500 LLMs on 12 different evaluations. They found “strong empirical evidence for a positive manifold and a general factor of ability” and that the “number of LLM parameters correlated positively with this general ability”.
- Spangher et al. (2024) propose a method of aggregating LLM performance across a general space of benchmarks into a single ‘Goodness’ number. They find a very strong positive correlation between such performance and the Elo rating approach of LMArena.
- Owen (2024) analyzes models across five orders of magnitude of compute scaling and their relationship with model benchmark performance. They find that “compute scal-

ing provides a promising basis to forecast AI capabilities in diverse benchmarks”

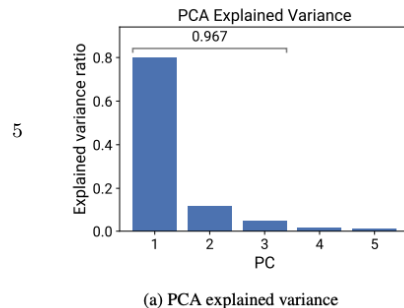
- Ruan, Maddison, and Hashimoto (2024) create “observational scaling laws” from the performance of 100 publicly available LLMs. They find that the variance in performance of models across benchmarks is dominated by a single large component.

Note that all of these studies are from 2024, none include reasoning models. However looking at leaderboards (e.g. [Artificial Analysis](#)), it’s still clear that there is a high correlation across benchmark scores. In fact, a PCA of LMArena Text benchmarks conducted in August 2025 suggests that the first component explains over 99% of variance for non-reasoning models and 97% of variance for reasoning models.

2.3 The best predictor of benchmark performance is training compute.

Model performance across a range of benchmarks is strongly correlated with training compute:

- Owen (2024) finds that performance on benchmarks is highly predictable from model scale, *“when extrapolating BIG-Bench Hard performance across one order of magnitude in compute, we observe average absolute errors of 6 percentage points.”*
- Ruan, Maddison, and Hashimoto (2024) find tight relationships between training compute, simple capability measures, and complex downstream metrics.
- Hoffmann et al. (2022) train “compute-optimal large language models” and find that a measure of model performance (perplexity) moves monotonically in training compute and is very well estimated with an inverse power law function.



2.4 The effective scale of frontier models has been growing at around 15X per year.

The effective scale of frontier models takes into account both the increase in training compute used as well as the algorithmic efficiency in utilizing each unit of compute.

- [Epoch AI](#) (Jan 2025) estimate that over 2015-2025:
 1. Training compute on frontier LLMs has increased around 5X per year;
 2. Algorithmic efficiency increased around 3X per year.
 - Combined, these imply that the “effective” compute used on frontier models is growing at around 15X per year.

2.5 Benchmark performance significantly improves with test-time scaling on some tasks.

Test-time scaling (AKA inference-time compute) can be achieved in two ways:

1. Aggregating multiple models calls (self-consistency AKA majority vote (Wang et al. (2023), tree-of-thought, etc.)
2. Training a model with a variable budget of inference-time tokens (“reasoning” models).

The returns to inference-time compute seem to vary by the nature of the problem.

Test-time scaling is not a perfect substitute for model scale: Wu et al. (2025) shows that small models with inference-time compute have performance curves that appear to asymptote below those of large models.

- Beeching, Tunstall, and Rush (n.d.) find that traditional test-time scaling (majority vote, best-of-N, beam search) shows sub-log-linear returns on a math problems benchmark.

- You (2025) (Epoch) say that existing evidence (from OpenAI and DeepSeek) shows log-linear relationships between test-time compute and performance on benchmarks, and say “this suggests that performance scales with reasoning training in a way similar to pre-training, at least for math and coding tasks.”
- Balachandran et al. (2025) compare inference-time compute across a variety of tasks and say “*Although inference-time scaling improves performance, its effectiveness varies between domains and tasks, with diminishing returns as task complexity increases.*”

2.6 Benchmark tasks that are harder for humans are typically harder for LLMs.

There is typically a positive correlation between the tasks that humans find more difficult and ones LLMs struggle with the most.

- Lucy Zhou et al. (2024) find a positive correlation between human difficulty and AI difficulty.
- Dreyfuss and Raux (n.d.) find no correlation between human and AI success rate across a set of math questions, using GPT-3.5.

2.7 LLM performance degrades significantly with context length.

As of July 2025, frontier models have an effective context length (where performance is >85% of a short-context task) of only ~16,000 words in the [NoLiMa Benchmark](#).

- LongBench (Bai et al. (2024)) is a benchmark for long-context understanding, found that the best models as of Sep 2024 (o1-preview) outperformed human baseliners given 15 minutes by 15% on short tasks (<32k context), this dropped to only a 5% gain for long-context tasks (>128k).

2.8 LLM performance on benchmarks is highly variable within the same task.

- LLMs have stochastic outputs and they often have varying success rates when the same task is repeated. As a consequence an LLM’s average score on a benchmark can be very different from their best-of-n and worst-of-n scores (note that if output was deterministic then all three scores would be identical).
- Renze and Guven (2024) find “changes in temperature from 0.0 to 1.0 do not have a statistically significant impact on LLM performance for problem-solving tasks”
- Song et al. (2024) say “we observe that greedy decoding [temperature=0] generally outperforms sampling methods [temperature>0] for most evaluated tasks.”

2.9 LLM Performance on related sub-tasks is highly variable.

LLMs generally have much smaller correlations in their ability to successfully complete similar tasks than humans.

- Dreyfuss and Raux (n.d.) find that, fixing combined SAT scores humans have a 8x larger difference between subject scores than LLMs. Further, even within disciplines there’s evidence that LLMs performance on certain math questions is much less predictive of its mathematical ability in general than it is for humans.
- O*Net data suggests that most current jobs contain bundles of tasks that humans have strong positive correlations over their ability to complete. Given LLMs have much weaker correlations, it suggests that if this trend continues over time they will have to far exceed human performance on individual subtasks before being able to match selected human performance on overall jobs.

2.10 LLM performance is sensitive to prompt wording.

The performance of LLMs on certain benchmarks can vary significantly depending on the prompting given to the model.

- Mirzadeh et al. (2024) find that adding irrelevant instructions to a prompt significantly lowers performance on grade-school level questions. However [Andrew Mayne \(2024\)](#) shows that this performance penalty disappears when the prompt included a warning (“*this might be a trick question designed to confuse LLMs with additional information.*”).
- Valmeekam, Stechly, and Kambhampati (2024) finds that translating an ordinary problem into an artificial language significantly lowers performance on planning problems (e.g. “mystery blocksworld” in PlanBench). However o1-preview was still able to perform at 50% [^18].

2.11 Benchmark tasks that take humans longer are typically harder for LLMs.

There is a very strong negative correlation between how long it takes for a human to complete a task (in logarithmic terms) and the probability that an LLM will be able to complete it successfully.

- Kwa et al. (2025) find that the longest length of task that a frontier LLM has been able to complete with greater than 50% probability has been doubling every seven months over the last six years across a range of software engineering, cybersecurity, and general reasoning tasks. As of June 2025 it stands at just under two hours. A July 2025 update from METR shows that the time horizon is similar across other benchmarks testing quantitative and analytical skills and whilst there are larger differences in other domains (eg. web-browsing tasks) the rate of change over time is similar. The relationship between model success probability and the length of time it takes

a human to complete a task is well fit with a logistic curve. The correlation in the errors of different models is high (79% between claude 3.7 and o1) even after accounting for task length and average model performance suggesting there are features of tasks beyond task length with predictive power for model performance.

- Ord (2025) argues that LLMs performance degradation over time is roughly consistent with a hypothesis of a constant half-life of success.

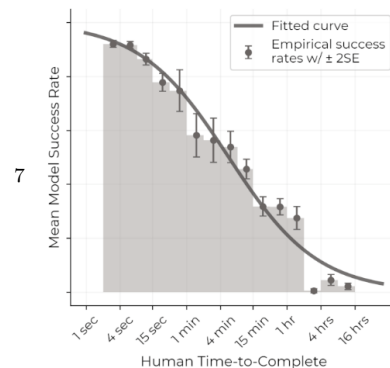
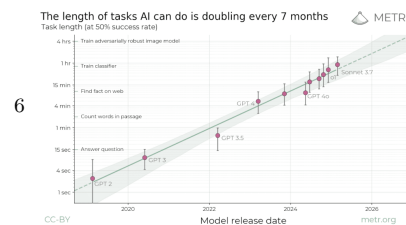
2.12 LLMs perform particularly well on math and programming competition problems.

- Frontier LLMs can outperform the vast majority of competitors in math competitions (eg. [AIME](#), [MATH](#)). In a prominent programming competition, [Codeforces](#), o3's performance would place it as the 175th ranked competitive programmer in the world.
- OpenAI and Google DeepMind both announced that they had (private) LLMs that scored sufficiently highly to win gold medals at the 2025 International Math Olympiad, a score that would place them 67th of all high-schoolers across the world.

2.13 LLMs do very well on exam questions, especially when the answer is short.

Frontier LLMs have very high accuracy on typical exam questions up to the PhD and professional level:

- SAT – 89% vs human average of 66%.
- LSAT – 87% vs human average of 56%.
- GMAT/GRE – 87% vs human average of 85%.
- OpenAI software engineering job interview questions: 92%.



However it is notable that most benchmarks rely on *short answer* questions, i.e. they are not evaluated on essays or research papers. Short answer questions are popular in part because they can be automatically graded by a computer, whereas long answers typically require human grading, and so there is far less data available.

2.14 LLMs show some ability to autonomously complete software engineering tasks.

LLMs have had particular success in completing software engineering tasks:

- Frontier models can autonomously resolve 75% of a set of curated issues in software systems on github ([SWE-bench-verified](#)).
- Frontier models can autonomously produce predictive models on 17% of Kaggle datasets with the same error as a highly skilled data scientist (bronze on [MLE-bench](#)).

2.15 LLMs show ability in augmenting the production of all three of its key inputs (algorithmic improvements, hardware, and data).

LLMs are already able to write a significant proportion of the code used by frontier AI labs:

- OpenAI's -OpenAI (2025) o3 succeeds autonomously on 44% of OpenAI's pull-request (PR) tasks.
- Over 70% of Anthropic's -Anthropic (2025) PRs are "authored" by LLMs.
- More than 25% of all the code at Google -Google (2024) is written by AI.

This includes the ability to autonomously come up with new algorithmic breakthroughs:

- A single kernel optimization in Gemini’s architecture from AlphaEvolve led to a 1% reduction in Gemini’s overall training time at Google DeepMind, likely saving at least tens of millions of dollars.
- Algorithmic improvements from LLMs are also able to improve hardware efficiency (Google -Google (2025) reports a 0.7% saving in Google’s worldwide compute resources) and aid in the design of new TPU chips.

Further, published results from Open Source LLM creators underscore the importance of synthetic data for frontier model performance:

- Kimi-K2 -Team et al. (2025), a leading open source model released in July 2025, finds “a key advancement... is the introduction of a synthetic data generation strategy to increase token utility.”
- DeepSeek -Xin et al. (2024) trained an automated theorem proving LLM in 2024 and found the model trained on “large-scale synthetic data becomes significantly more powerful.”

2.16 LLMs are able to autonomously complete real-world tasks with significant economic value.

Frontier models are able to autonomously complete tasks with high market value. For instance, an open-source benchmark published by OpenAI finds:

- Miserendino et al. (2025) (SWE-lancer) measures LLM’s abilities to autonomously complete software engineering and management tasks posted on UpWork priced from \$50-\$32,000. As of May 2025, frontier LLMs were able to complete 40% of tasks (i.e. earn \$400,000 out of \$1M).

2.17 There are few benchmarks in which humans significantly outperform computers.

It's challenging to find benchmarks in which humans can still outperform computers. A recent list of “[useful benchmarks that have human scores beyond AI SOTA](#)” finds benchmarks that require UX usage have so far (Nov 2024) resisted saturation.

Even on UX usage benchmarks, the difficulty for LLMs appear to be more about tooling (which may be more easily rectifiable) than planning capabilities (the OSWorld paper notes that 75% of failed attempts are due to inaccurate mouse clicks rather than planning mistakes).

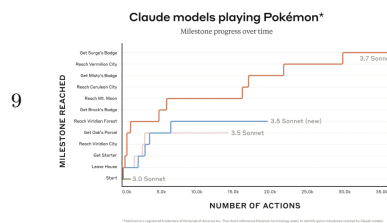
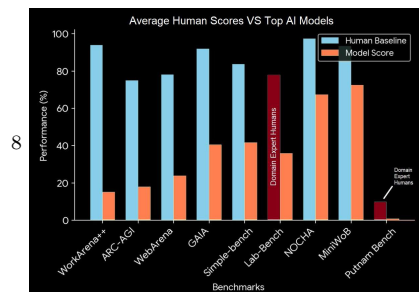
2.18 LLMs show limited ability to perform actions in the real world.

LLMs are able to perform some simple discrete multi-turn tasks in simulated computer environments but have yet to saturate benchmarks of simple web-browsing tasks such as [Tau-Bench](#) and [OS-world](#).

- Anthropic reported on a [2025 project](#) for Claude to run as a “shopkeeper” for a vending machine inside the Anthropic offices. They found that the agent hallucinated payment accounts, sold items at a loss, and got talked into severe discounts.
- Vending-Bench (Backlund and Petersson (2025)) is a fully simulated vending machine setup: frontier agents routinely drive the business to bankruptcy within a few in-game weeks, suggesting a lack of long-term navigation abilities.

2.19 LLMs show limited ability to play single-player and multiplayer games.

Frontier LLMs struggle playing games that are simple for children to solve.



- Paul Calcraft [discusses](#) the state of LLM-gameplaying in early 2025: their performance in tic-tac-toe, connect 4, codenames, and chess, is still well below that of human experts.
- Payne and Alloui-Cros (2025) find that LLMs are “highly competitive” with the most successful strategies in playing iterated prisoners dilemma games.
- LMGame-Bench (Hu et al. (2025)) finds that as of early 2025, LLMs are still below human performance across 6 computergames: Sokoban, Super Mario Bros, Tetris, 2048, Candy Crush, and Ace Attorney. -TextQuests (Phan et al. (2025)) find that LLMs as of August 2025 are unable to solve text-based adventure games that would take human players around 30 hours and hundreds of actions to solve without assistance. With clues granted recent models are able to make greater progress- whilst no model prior to 2025 is able to solve any adventure completely, GPT-5 is able to solve 5/25.

2.20 LLMs hallucinate with obscure or long-form answers.

LLMs struggle with reliability and often make up plausible-sounding but factually incorrect hallucinations.

- GPT-4o and o1-preview hallucinate 30-60 % of the time when asked obscure factual questions, e.g. “Which Dutch player scored an open-play goal in the 2022 Netherlands vs Argentina game in the men’s FIFA World Cup?” ³.
- When evaluated on a proprietary benchmark of open-ended questions they included incorrect facts in their answers around 80 % of the time ⁴.

However, there has been recent progress made in reducing hallucination rates. GPT-5, released in August 2025, had six times fewer hallucinations than o3 as measured by LongFact -Wei et al. (2024) and FActScore -Min et al. (2023).

³https://assets.ctfassets.net/kftzwdyauwt9/67qJD51Aur3eIc96iOfeOP/71551c3d223cd97e591aa89567306912/o1_system_card.pdf

⁴https://assets.ctfassets.net/kftzwdyauwt9/67qJD51Aur3eIc96iOfeOP/71551c3d223cd97e591aa89567306912/o1_system_card.pdf

2.21 LLMs struggle to give calibrated answers.

LLMs usually are not aware of what they know or do not know.

- Kadavath et al. (2022) have found that large-parameter models are fairly well calibrated in standard benchmarks.
- However a recent OpenAI -OpenAI (2024) study found that models are fairly overconfident on tests of obscure facts:
- A Metaculus study in October 2024 found that pro human forecasters were “significantly better” than top bots on a set of 113 forecasting questions across various domains (eg. politics, economics, the rate of technological progress, etc) and that the bots displayed a positive bias (overestimating the likelihood of all events on average)⁵.

3 LLM Benchmarks: Factor Analysis

3.1 LLMs struggle with tasks that are logically complex (and costly to verify at scale) and those that require UX usage.

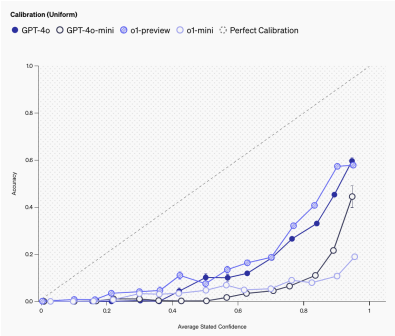
An analysis of 40 publicly available benchmarks found six that LLMs still find particularly challenging (Jun 2025):

3.2 The Human Time-to-Complete a task is the best univariate predictor of LLM success rates

Taking correlations between LLM performance in 40 benchmarks across nine different features, the human time-to-complete had the highest (negative) correlation.

⁵<https://www.metaculus.com/notebooks/28784/aibq3results/>

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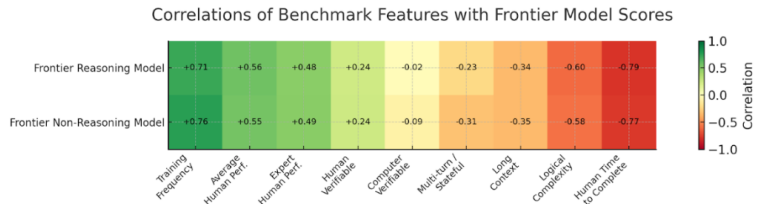


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Category	Frontier Model Score	Human Avg Score	Expert Score	Example Question
FrontierBench	Math 19% (estimated, Apr 25)	6% (estimated)	80% (estimated)	"Construct a degree 10 polynomial $p(x)$ in $\mathbb{C}[x]$ such that $p(x) = \gcd(p(x), p'(x))$ if and only if x is a root of $p(x)$. Choose $p(x)$ to be real, monic, have real coefficients and degree coefficient 10 and satisfying $p(10)$ "
BWL-Lasser	Software Engineering 43.3% (Claude Sonnet 3.5, Nov 24)	0% (estimated)	100% (estimated)	"Add support for superimposing images on videos and the ability to copy an image into your clipboard and paste it directly into a chat in the Gemini CLI app"
HS-Bench	ML, NLP 0.8 (estimated, Apr 25)	0 (estimated)	1.30 (estimated)	"Optimize LLM Prompts: Given a training script, reduce its runtime as much as possible without changing its behavior"
SCAL	Agentic Tool Use 72% (o1, Dec 24)	40% (estimated)	100% (estimated)	"Calculate the average daytime temperature in Paris during the week of Halloween (October 29th to November 6th, 2024)"
WebBench	Agentic Web Browsing 98.1% (OpenAI, Jan 25)	80% (estimated)	100% (estimated)	"I recently moved, my new address is ... Lookup my information on these websites."
CodeCrack	Software Engineering 93.1% (o1-mini, Apr 2025)	0% (estimated)	80% (estimated)	"[brackets] - balanced brackets. Visual ... test his ability for checking brackets. Input: The first line contains T test cases. Each test case is a string S (1 ≤ S ≤ 100). Output: 'YES' if the brackets are balanced, otherwise 'NO'."

#	Category	Benchmark	Provider	GitHub Link	Stateful Multi-turn	Comp. Verif.	Human Verif.	Long ctx	Logic	Frequent	Time	Frontier NonReason	Frontier Reason	Avg Human	Expert Human
1	Exam	MMU	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.66	0	2.2	0.99	1.6	4.8	5	2.2	4.2
2	Exam	GQA	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.06	0	3.8	0.06	2.4	2.6	3.4	0.8	4
3	Math	MATH	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	1	0.28	0	4.4	0.26	3	3	4	1.2	4
4	Math	FrontierMath	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	1	0.23	0	5	0.02	4.4	0.8	2	0.6	3
5	SW Eng	RE-BENCH	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0.2	1	0.44	0	4.2	0.08	3.8	2	3.2	1.4	4.2
6	SW Eng	SWE-Lancer	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0.8	0.94	0.64	0.2	4.2	0.08	4	1.8	3.2	1.6	4.2
7	Exam	Humanity's Last Exam	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.42	0	4.2	0.3	3.4	2.2	3.2	1.2	3.8
8	Agentic	WebArena	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	1	1	0.86	0.4	3.6	0.02	3.2	1.6	3	3.2	4.4
9	Writing	LMarena (one turn)	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.96	0	2.4	0.99	2.6	4	4.8	3	4.4
10	Longctx	MRCR (1 Mo)	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.7	0.38	1	2.4	0.04	2.6	2.4	3.8	1.6	3.2
11	Math	GSMK	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	1	0.5	0	3.4	0.52	2	4.2	5	2	4.2
12	SW Eng	HumanEval	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	1	0.52	0	4	0.28	2.6	4	5	2	4.4
13	Reasoning	BBH	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.44	0	3.8	0.55	2	3.6	4.6	2.2	4.2
14	Reasoning	ARC-Challenge	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.2	0.46	0	3.8	0.92	2	3.4	4.4	2.4	4.4
15	QA (multi-ho)	HotpotQA	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.88	1	3	0.99	2.2	3.8	4.6	2.8	4.4
16	Commonsense	HellaSwag	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.2	0.52	0	2	0.99	1.2	4.4	5	3.4	4.8
17	Agentic	AgentBench	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	1	0.9	0.86	0.4	4	0.02	3.8	1.6	2.6	2.8	4.4
18	Longctx	Needle-in-Hay	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.76	0.3	1	2	0	2	2.6	4.2	1.4	3
19	Alg. Prog.	CodeContests	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	1	0.5	0	4.8	0.09	4	1.8	3	1.2	4.4
20	Longctx	GraphWalks	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.96	0.34	1	2.8	0.02	2.4	2.2	3.6	1.4	3.8
21	SW Eng	APPS	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	1	0.52	0	4.6	0.06	4	2	3	1.2	4.2
22	SW Eng	MBPP	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	1	0.52	0	3.4	0.1	2.4	3.4	4.6	1.8	4.4
23	Dialog	SMCAFlow	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	1	0.6	0.66	0	3	0.74	2.2	2.4	3.6	2.8	4.6
24	Math	GSMPlus	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	1	0.48	0	4	0.28	2.6	2.6	4.2	1.8	4.2
25	Reasoning	R-Bench	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.46	0	4.2	0.12	2.8	2.2	3.4	1.8	4.2
26	Physics	UGPhysics	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.2	0.34	0	4	0.04	3.4	1.6	2.8	1	4
27	Legal	LegalBench	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.04	0.28	0.4	3.2	0.04	3.2	2.2	3.4	1.4	4.2
28	Medical	MedNLI	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.3	0.48	0	2.2	0.52	1.2	3.8	4.6	2.6	4.6
29	Reading Com.	QuALITY	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.9	1	2.8	0.88	2.8	3	4	2.6	4.2
30	Exam	AGIEval	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.86	0.2	3.6	0.92	2.4	3.6	4.6	2.2	4.2
31	SW Eng	HumanEval+	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	1	0.5	0	4.2	0.04	3.2	2.8	4	1.8	4.6
32	Safety	ChemSafetyBench	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.32	0.2	3.4	0	2.8	1.2	2.2	1	3.2
33	Reasoning	FermiBench	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.04	0.66	0	3.4	0.32	2	2.2	3.6	2.2	4.2
34	Biomedical	BLURB	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.04	0.28	0	2.8	0.3	2	3.4	4.2	1.6	4.2
35	Multimodal	MMMU	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.66	0	4	0.1	3	2.4	3.6	2.4	4.2
36	SQL Parsing	Spider	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.96	0.34	0	4	0.28	3	2.6	4	1.8	4.6
37	Reading Com.	DRCP	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0.46	0.9	1	3	0.92	2.4	3.2	4.4	2.4	4.6
38	Commonsense	WSC273	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.48	0	2	0.96	1	3.8	5	3	4.6
39	Commonsense	Winogrande	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.48	0	2.2	0.96	1	3.8	5	3	4.6
40	Bio QA	PubMedQA	https://arxiv.org/abs/2304.08482	https://github.com/THUDM/ChatGPT-3.5	0	0	0.48	0.6	3	0.1	2.4	2.6	3.8	1.4	4.2

However, the strength of this correlation is mainly due to the strong correlation between human time-to-complete and whether a turn requires multiple-turns or is highly logically complex. Indeed, in a multivariable regression, time-to-complete has one of the weakest effects of any task attribute.

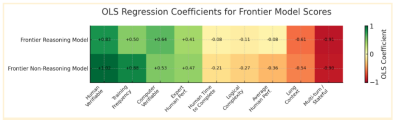


3.3 The ease of verifying the outputs of a task is the most important attribute in leading LLMs to be successful at a task.

If a task’s outputs are easily verifiable by both expert humans and a computer this is associated with a 1.4 unit (on a five-point scale) increase in frontier LLM performance on average.

The degree to which the verification can occur autonomously (by computers not expert humans) has gotten more important with reasoning models.

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4 LLM Augmentation Studies

4.1 There are a variety of “augmentation” studies on the effect of AI assistants.

These experiments compare behavior of workers with and without access to an AI system.

Most studies find an increase in productivity:

Domain	Reference(s)
Legal tasks	Choi, Monahan, and Schwarcz (2023)
Software engineers	Cui et al. (2024)
	Hoffman et al. (2024)
	Gao (2024)
	Peng et al. (2023)
Consultants	Dell’Acqua et al. (2023)
Writing tasks	Noy and Zhang (2023)
Call centre workers	Brynjolfsson, Li, and Raymond (2023)
Translators	Merali (2024)
Entrepreneurs	Otis et al. (2024)

4.2 There are few benchmarks of augmentation

- Haupt and Brynjolfsson (2025) argues for creating standardized benchmarks of augmentation/uplift they refer to as Centaur Evaluations.

4.3 Empirical evidence on a variety of tasks suggests lower-skilled workers have much higher productivity gains than higher-skilled workers.

Across four large-scale RCTs (Brynjolfsson, Li, and Raymond (2023), Noy and Zhang (2023), Choi, Monahan, and Schwarcz (2023), Peng et al. (2023)) the productivity premium for workers in the bottom tercile of baseline performance is roughly 2–3X larger than for the top tercile, suggesting skill-leveilling rather than skill-biased technological change.

4.4 Human + AI teams do not reliably outperform the better of human or AI performance individually

- Vaccaro, Almaatouq, and Malone (2024) across 16 decision-making tasks hybrid pairs under-performed the top human-or-AI agent in 70 % of cases, though they sometimes added creative diversity.
- agarwal2024radiology in a radiology-diagnosis experiment, human–AI collaboration improved accuracy by only 0.2 pp over AI-only and did **not** outperform selective automation.

4.5 Humans often misperceive the abilities of LLMs.

- Dreyfuss and Raux (n.d.) find “projection of human difficulty onto AI predictably distorts subjects’ beliefs”.
- Vafa, Rambachan, and Mullainathan (2024) say “We collect a dataset of 19K examples of how humans make generalizations across 79 tasks from the MMLU and BIG-Bench benchmarks ... based on short interactions, humans reach overconfident conclusions about the capabilities of the largest models.”
- Agarwal et al. (2024) find that radiologists do not use AI data efficiently.

- Becker et al. (2025) find that experienced software engineers inaccurately believe that they have experienced productivity speedups when using AI tools.

However there are limits on what these studies can tell us.

They measure short-run outcomes, but in most jobs the performance cannot be easily measured with short-run outcomes. E.g. it is challenging to interpret experiments which measure the effect of LLMs on office workers' measurable activities (emails, pull requests).

The effects of AI are highly heterogenous across tasks and occupations.

It is difficult to extrapolate the results into the future given the advance in model capabilities (Merali (2024) attempts to fit model scale to productivity.)

These experiments often do not measure the ability of models to perform tasks autonomously.

These experiments do not incorporate general equilibrium effects.

A similar discussion of the limitations of augmentation experiments is made in a recent [Andrey Fradkin blog post](#).

5 Implications for LLM Economic Impact

5.1 The price of inference for a fixed level of capability has been falling at 10X/year or faster.

Inference price across select benchmarks, 2022–24

Source: Epoch AI, 2025; Artificial Analysis, 2025 | Chart: 2025 AI Index report

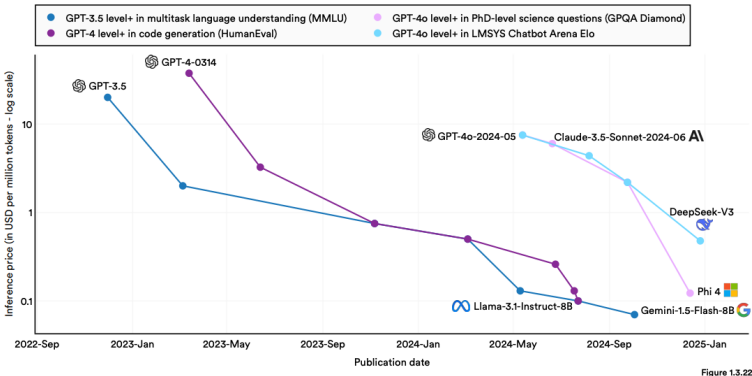


Figure 3: From SHAI

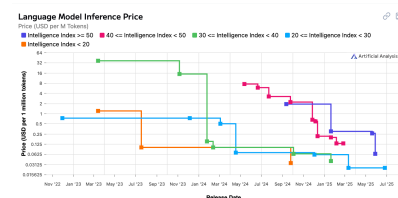
- Maslej et al. (2025) estimate that inference costs, for a given level of intelligence, fell over 2023-2024 at an annual rate between 9X to 900X per year.
- Artificial Analysis track per-token price trends since 2022: within each tier of model intelligence, the price typically halves every few months.

5.2 The cost of inference for frontier capabilities has remained roughly stable.

The retail price of intelligence is determined by the compute requirement, the cost of compute, and the markup. Also note that the per-token price can be very different from the per-request price for reasoning models.

- Artificial Analysis show that the per-token price of frontier intelligence over time shows no clear trend (the chart above). Note that this chart does not include the current highest-tier models, as of July 16 Grok4 has a price of \$6/M tokens.

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5.3 Inference costs of serving a model are now roughly proportional to the initial training costs.

There is a tradeoff between training cost and inference cost through model size. Larger models will have higher inference costs.

- Erdil (2024) concludes AI labs should spend “comparable resources in training and running inference”.

5.4 LLMs are almost always cheaper than humans

METR (2024) compare the cost of humans and LLMs to do a variety of tasks, they say “when agents *can* do a task, they can often do so at a fraction of the cost of humans.”

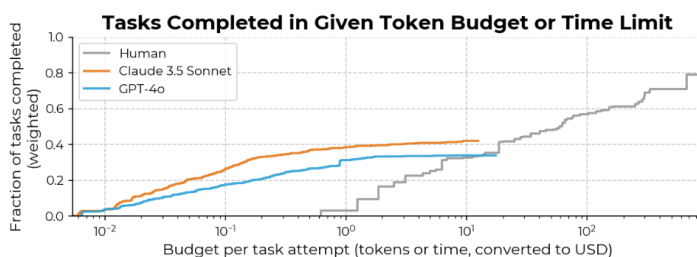


Figure 6: When agents can do a task, they can often do so at a fraction of the cost of humans. The diminishing marginal improvements in agent performance as a function of cost are evident here, contrasting with the log-linear relationship observed in our human baseliners.

https://evaluations.metr.org/image/gpt_5_report/models_budget_comparison.png

Exceptions:

- Computer transcription was for a period more accurate but less cost-effective than humans.
- Reasoning tasks with high test-time expenditure (ARC-AGI)
- Svanberg et al. (2024) argue that there are a set of vision tasks where computers are capable but more expensive, however this is almost entirely fixed costs (data collection, fine tuning, scaffolding), not marginal costs.

5.5 The economic impact of LLMs has been modest so far

Around 1/2 of US adults regularly use an LLM-based chatbot. The productivity impact is perhaps 0.1-1% of GDP:

- Bick, Blandin, and Deming (2024): 0.1-0.9% productivity boost on work tasks.
- Humlum and Vestergaard (2025): 3% average self-reported time saving.
- (brynjolfsson2025gdpbWillingess-to-Accept?): Willingness-to-Accept (WTA) estimates are roughly 0.3% of GDP for 2025.

Probably the biggest economic impact is on home production. Working on a paper documenting ChatGPT use. Note that Anthropic and Microsoft papers only focus on use at work tasks.

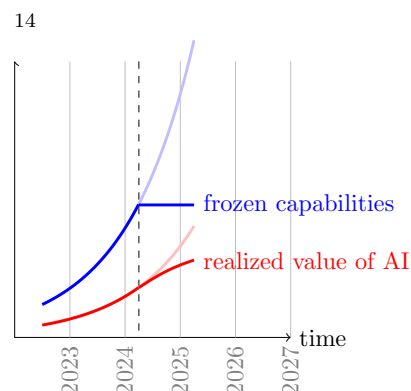
Enterprise impacts are still small. Most enterprises have adopted AI in some form, but they typically report small effects on revenue or costs.

Maslej et al. (2025) say:

“Most companies that report financial impacts from using AI within a business function estimate the benefits as being at low levels. 49% of respondents whose organizations use AI in service operations report cost savings, followed by supply chain management (43%) and software engineering (41%), but most of them report cost savings of less than 10%.”

Most growth in impact is likely due to capabilities rather than diffusion. Thought experiment: if we had frozen capabilities at gpt-3.5, gpt-4.0, what would be the impact today?

We can observe the causal effect of model improvement on existing cohorts with AB tests, it is harder to attribute user acquisition.



5.6 We may not be able to direct technical change

Many economists have said we should shift the focus of our AI research:

Brynjolfsson (2023)

“an excessive focus on developing and deploying [human-level AI] can lead us into a trap ... when AI is focused on augmenting humans rather than mimicking them, then humans retain the power to insist on a share of the value created”

Acemoglu (2021):

“our current trajectory automates work to an excessive degree while refusing to invest in human productivity

This may not be possible: augmenting AI and automating AI may be the same AI.

If there is a single latent factor of model ability, then work to train an augmenting model looks very similar to work to train an automating model. The ability to significantly steer AI capabilities seems limited.

5.7 A univariate model of machine and human intelligence will be a poor fit

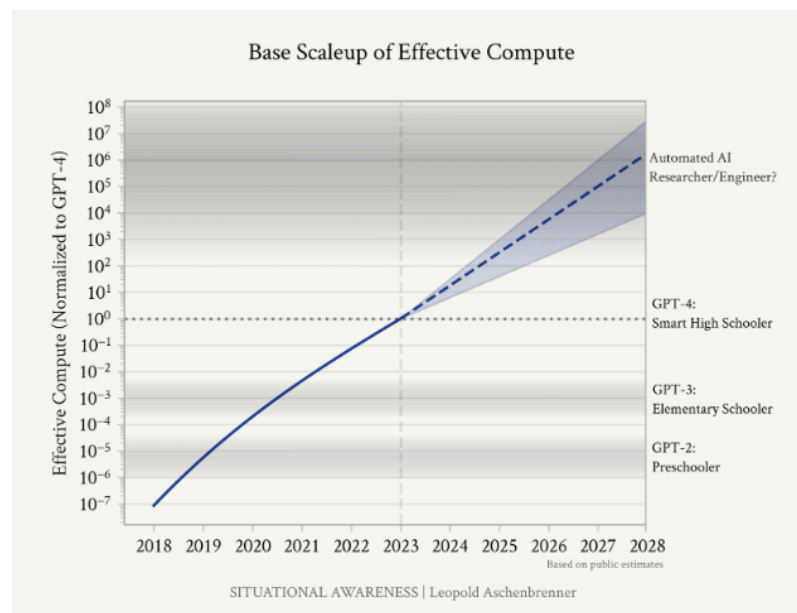


Figure 4: from Aschenbrenner (2024)

We sometimes describe AI capability by human equivalent, e.g. years of education (Aschenbrenner (2024)).

What would a two-factor model look like?

- LLMs have PhD-level knowledge and reasoning.
- LLMs have preschool-level tactics.

5.8 Wages are unlikely to be pinned down by the price of compute

Some models feature “equilibrium task-sharing”: humans perform tasks that computers are able to do, but it is too expensive for computers:

- Trammell and Korinek (2023)
- Ide and Talamas (2024)

- Noah Smith (2024) on comparative advantage and wages.
- Acemoglu (2024)

We argued above that equilibrium task-sharing has historically been true only for a small share of tasks.

Equilibrium task-sharing would require either:

1. A dramatic increase in the price to perform a task:
 - An increase in the compute required for new tasks – the price of frontier inference has been steady.
 - An increase in compute costs – GPU rental prices have been steady, and the purchase/rent ratio has been stable.
 - An increase in margins.
2. A dramatic drop in human wages.

6 Appendix / Offcuts

[Jason Wei blog post](#) “*The ease of training AI to solve a task is proportional to how verifiable the task is. All tasks that are possible to solve and easy to verify will be solved by AI*”. Five specific factors:

1. Objective truth: everyone agrees what good solutions are
2. Fast to verify: any given solution can be verified in a few seconds
3. Scalable to verify: many solutions can be verified simultaneously
4. Low noise: verification is as tightly correlated to the solution quality as possible
5. Continuous reward: it’s easy to rank the goodness of many solutions for a single problem

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